Sara Vannah

Atmospheric and Environmental Research (AER)
JANUS Research

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Education

Dartmouth College Feb. 2023
PhD in Physics & Astronomy *Advisor: Marcelo Gleiser*

Wellesley College May 2017
Bachelors of Arts in Astrophysics *Advisor: Glenn Stark*

Research Experience

Senior Research Associate AER JANUS Research
Winter 2023 — Present

Developed patch for radio occultation processing tool to accept data from AWS Open Data registry. Compared refractivity retrievals between different processing centers. Performed air quality analysis projects for Texas Commission on Environmental Quality. Assessed the progression of the widths of the Tropics over decades through the locations of the Hadley cells in various models. Compared observations on models of Great Plains low-level jet to assess their ability to capture the physics of the jet.

Doctoral Research Dartmouth College
Advisor: Marcelo Gleiser *Fall 2017 — Winter 2023*

Developed information-theory based methods for analyzing cosmic microwave background and exoplanetary atmosphere spectra. Wrote information theory-based Markov Chain Monte Carlo for analyzing the cosmic microwave background. Mentored two undergraduate research projects

Student Researcher Wellesley College
Advisor: Glenn Stark *Spring & Fall 2016*

Demonstrated strong self-shielding effects in carbon monoxide. Performed reduction of sulfur dioxide spectroscopy data in search of a quantum-mechanically “forbidden” transition

KNAC Research Fellow Vassar College through KNAC REU Program
Advisor: Colette Salyk *Summer 2016*

Composed Python routines to perform spectral analysis on a series of protoplanetary disks. Used analysis to determine properties of the disks

Research Intern Louisiana State University CCT REU Program
Advisor: Juhan Frank *Summer 2015*

Developed an efficient, simplified analytic simulation of mass transfer in double white dwarf systems to supplement computationally expensive numerical simulations.

Publications

1. S.S. Leroy, **S. Vannah**, S. Ming, & P. Lin. (2026) "Diagnostics of the Great Plains low-level jet in atmospheric reanalyses", *in review in Journal of Applied Meteorology and Climatology*.

2. S.S. Leroy & **S. Vannah**. (2026) "Widening of the tropics in global surface-air winds", Nature Scientific Reports *16*, 12344. [doi:10.1038/s41598-026-43234-z]

3. **S. Vannah**, I. Stiehl, & M. Gleiser. (2025) "An Informational–Entropic Approach to Exoplanet Characterization", Entropy *27*(4), 385. [doi:10.3390/e27040385]

4. **S. Vannah**, S.S. Leroy, C. O Ao, E. R. Kursinski, K.J. Nelson, K.-N. Wang, & F. Xie. (2025) "The impact of differences in retrieval algorithms between processing centers on GNSS radio occultation refractivity retrievals in the planetary boundary layer", Atmospheric Measurement Techniques *18*, 4293–4310. [doi:10.5194/amt-18-4293-2025]

5. **S. Vannah**, M. Gleiser, & L. Kaltenegger. (2024) "An information theory approach to identifying signs of life on transiting planets", Monthly Notices of the Royal Astronomical Society: Letters *528*(1), L4-L9. [doi:10.1093/mnrasl/slad156]

6. V. May, R. Sloboda, M. Tine, S. Streeter, D. Clemens-Sewall, **S. Vannah**, G. Goebel. (2022) "Feast or Famine Terrarium Project", Paper presented at 2022 ASEE Annual Conference & Exposition, Minneapolis, MN https://peer.asee.org/40723

7. M. Stephens, **S. Vannah**, & M. Gleiser. (2020) "Informational Approach to Cosmological Parameter Investigation" Physical Review D **102**, 123514 [doi:10.1103/PhysRevD.102.123514]

8. **S. Vannah** & C. Salyk. (2016) "A CO Spectral Analysis of Protoplanetary Disks", Keck Northeast Astronomy Consortium Symposium 2016, Wesleyan University, Middletown, CT. Paper published in conference proceedings

Posters & Presentations

*denotes invited talk

Information theory tools in the the search for life on exoplanets
Boston Area Planetary Sciences Meeting, Cambridge, MA (2025).

Quality Control in the Planetary Boundary Layer: An Intercomparison of GNSS Radio Occultation Retrievals Processed by UCAR, JPL, and ROM SAF
International Radio Occultation Working Group-10, Boulder, CO (2024) *talk* slides

An Information Theory Framework for Identifying Signs of Life on Transiting Exoplanets
Astrobiology Science Conference, Providence, RI (2024) *talk** abstract

Information theory as a tool in the search for life on exoplanets
MIT Planetary Lunch Seminars, MIT (2024) *talk*

Information Theory and the Search for Life in the Universe
Yuk Li Yung Lunch Seminar, CalTech (2024) *talk**

Comparing the Evolution of Earth-like Exoplanets Orbiting FGKM Stars: An Information Theory Approach
Planetery Science Informatics and Data Analytics, Villa de Cañada, Spain (2022) *talk*
slides, recording

A CO Spectral Analysis of Protoplanetary Disks
229th meeting of the American Astronomical Society, Grapevine, TX. (2017) *poster*
abstract

Ancestral Astronomy: A Spectral Analysis of Protoplanetary Disks
Tanner Conference, Wellesley College, Wellesley, MA. (2016) *talk*

A CO Spectral Analysis of Protoplanetary Disks
Keck Northeast Astronomy Consortium Symposium, Wesleyan University, Middletown, CT. (2016) *talk*

Simplified Simulation of Mass Transfer in Double White Dwarf Systems
227th meeting of the American Astronomical Society, Kissimmee, FL. (2016) *poster*
abstract

Awards & Honors

JANUS Energy Award
AER JANUS Research — *2026*

International Travel Grant
American Astronomical Society — *2022*

Selamawit Tsehaye Teaching Award
Dartmouth College — *2020*

Nominated for Guarini Teaching Award
Dartmouth College — *2020*

Dartmouth DCAL Outstanding Graduate Student Teacher Award
Dartmouth College — *2018*

New Hampshire Space Grant
NASA — *2017*

Sigma Xi research honor society nominee
Wellesley College — *2017*

Teaching, Service, & Outreach

Teaching Assistant — Winter 2018 — Fall 2022
Dartmouth College — *Hanover, NH*
- Led labs for nearly a dozen physics and astronomy courses, primarily introductory mechanics and electromagnetism
- Won both student and faculty-nominated awards for teaching
- Designed and led recitations
- Graded assignments quickly and effectively

Graduate Science Partner — Sept, 2021 – May, 2022
Darmouth Rural STEM Educator Partnernship — *Hanover, NH*
- Developed publishable lesson plans for a hands-on science curriculum for middle school students using Arduinos and terraria
- Met with middle school teachers, used feedback to adjust lessons to better fit rural students' needs
- Filmed and edited video versions of lessons for low-literacy students
- Visited local middle schools to interact with students, help implement lessons

Graduate Mentor — Spring 2021
Dartmouth Many Mentors — *Hanover, NH*
- Served as science mentor for rural Nevada high school

Public observing volunteer — 2015-2017
Wellesley College — *Wellesley, MA*
- Led hands-on lessons for young children using the college's meteorite collection

Student Physics Department Representative — Fall 2016-Spring 2017
Wellesley College — *Wellesley, MA*

- Served as the student representative in department meetings
- Successfully changed seminar times to allow students with afternoon work-study jobs and athletes to attend

Society of Physics Students Treasurer Fall 2016-Spring 2017
Wellesley College *Wellesley, MA*
- Managed the finances for the club
- Helped organize and lead activities including both public outreach and activities within the department to help create a sense of belonging in physics

Specialized Skills

Programming Languages: Python (advanced), Mathematica (intermediate), MATLAB (beginner), Java (beginner)
Computational skills: object-oriented programming, data structures, machine learning, algorithms, high-performance computing
Technical skills: statistics, signal processing

Professional training

Astrostatics bootcamp at Pennsylvania State University, 2021

Future Faculty teacher training, Dartmouth Center for the Advancement of Learning, 2020

Scientific Communication course, Dartmouth College, 2018